Capstone Documentation: List of Fixes

Source: Capstone documentation.pdf. The system (code) is correct; the document should be changed to match it. Total: 96 items (42 High, 38 Medium, 16 Low).

1. Chapters 1-3 (22)

#	Severity	Where in document	Problem and fix
1	High	Tab 1 / Specific Problem (Librarian, line 173-174) & Scope (Librarian, line 791)	Remove "audit logs" from Librarian dashboard shortcuts and specific problems; state that audit logs are strictly restricted to Super Admin.
2	High	Tab 1 / Scope and Limitations / Limitation (line 1009-1010)	Remove "and local thesis soft copies" from the catalog limitation; state that the system exclusively manages physical book assets.
3	High	Tab 1 / Scope and Limitations / Limitation (line 1001-1002)	Revise text to state that accession numbers are auto-generated sequentially by the database (`ACC-XXXXX`), while Dewey Decimal classification can be retrieved via ISBN API lookup or manually specified.
4	High	Tab 1 / Specific Problem (SA, line 255-259), Objectives (SA, line 558-561), Scope (SA, line 869-873)	Update description to clarify that the Student Assistant dashboard displays their personal digital library card, personal activity logs, and an Assistant Desk Shortcuts toolbar (when permissions are granted), rather than system-wide metric stat cards.
5	High	Tab 1 / Specific Problem (SA, line 268-274), Objectives (SA, line 569-574), Scope (SA, line 883-890)	Update text from "structured data table" / "tabular overview" to "responsive card grid catalog", and specify that Student Assistants have view-only access while batch tools and book creation require Super Admin or Librarian privileges.
6	High	Tab 1 / Specific Problem (Super Admin line 111-112, Librarian line 204), Scope (Super Admin line 730-731, Librarian line 823-824)	Update description to state the donut chart visualizes book collection distribution by category, and mention the peak hours bar chart for visitor traffic.
7	High	Tab 1 / Scope and Limitations / Student / Notification Module (line 985-986)	Correct text to state that the notifications page allows students to browse, filter, search, and manage system announcements, loan reminders, and hold alerts.
8	High	Tab 1 / Objectives (Librarian Analytics line 508), Scope (Librarian Inventory line 796, Librarian Analytics line 819)	Replace "Super Administrator" with "Librarian" in lines 508, 796, and 819.
9	High	Tab 1 / Scope and Limitations / Student / Category Module (line 960)	Change "Student Assistant" to "Students" in line 960.
10	High	Tab 1 / Objectives (SA Attendance lines 597, 600) and Scope (SA Attendance line 914)	Change "students" to "Student Assistants" in lines 597, 600, and 914 to preserve section role integrity.
11	Med	Tab 1 / Table 4 Minimum Hardware Requirements (For Users) (line 1486)	Replace "Integrated sound card (100% DirectX 10 compatible)" with standard display adapter (1024x768 resolution minimum) and a modern web browser (Chrome, Edge, Firefox, Safari) with web camera support for QR scanning.
12	Med	Tab 1 / Table 3 Minimum Hardware Requirements (For Development) (lines 1467-1471)	Reformat Table 3 columns cleanly: fix `3.6 GHz`, AMD Ryzen 3 3200G (4 cores, 4 threads), place `AMD Ryzen 5 5600X @ 3.7GHz (6 cores, 12 threads)` entirely in the Processor row, and place `8GB DDR4` (Min) / `16GB DDR4` (Rec) cleanly in the Memory row.
13	Med	Tab 1 / Table 2 Software Requirements (lines 1453-1457)	Update Database to "Supabase (PostgreSQL 15+)"; update technologies to "Next.js 16, React 19, TypeScript 5, Tailwind CSS"; correct "Microsoft 0365 Word" to "Microsoft Office 365 Word".
14	Med	Tab 1 / Technical Background (lines 1288-1293 and lines 1310-1316)	Delete the duplicate entry at lines 1288-1293, merge accurate text into lines 1310-1316, and remove claims of OTP generation.
15	Med	Tab 1 / Objectives / Student Login Module (line 628)	Change "multi-step profile completion workflow" to "single-step onboarding profile completion form".

#	Severity	Where in document	Problem and fix
16	Med	Tab 1 / Figure Captions / Figure 2 (line 1526) and Figure 6.6 (line 1593)	Update Figure 2 caption to "Figure 2 Data Flow Diagram Level 0" and Figure 6.6 caption to "Figure 6.6 Student-Assistant Borrowing Logs Flowchart".
17	Med	Tab 1 / Methodology / Diagrams (lines 1403-1405) vs Figure Captions (lines 1519-1607)	Add the Use Case Diagram figure number and caption into the methodology sequence, or adjust narrative if the Use Case Diagram was merged or omitted.
18	Low	Tab 1 / Specific Problem / Super Admin Dashboard (lines 81-83)	Delete ".real time" at the end of line 82.
19	Low	Tab 1 / Specific Problem / Super Admin Audit Logs (line 143)	Add a period at the end of line 143.
20	Low	Tab 1 / Scope and Limitations / Super Admin (lines 733, 744, 749, 762)	Correct to "User Directory Module", "super administrator", "Settings and Policies Module", and "Super Administrator".
21	Low	Tab 1 / Scope and Limitations / Student Assistant Dashboard (line 872-873)	Correct to "...showing their current borrows, active reservations, and recent attendance history."
22	Low	Tab 1 / Scope and Limitations / Limitation (line 1013)	Add a period at the end of line 1013.

2. Requirements (Tab 2) (12)

#	Severity	Where in document	Problem and fix
1	High	Tab 2 / Section 4.1.14 / REQ042	Remove "Audit Logs" from REQ042. Update to: "The librarian shall be able to use quick navigation shortcuts to access the User Directory and Analytics, as well as review recently cataloged assets."
2	High	Tab 2 / Section 4.1.25 / REQ053	Rewrite REQ053: "The student assistant dashboard shall display the digital Library Card, Assistant Desk Shortcuts (for permitted modules), personal activity tracking, and Support FAQs, without administrative summary metrics."
3	High	Tab 2 / Section 4.1.6 & 4.1.18 / REQ020, REQ046	Replace "donut chart breaking down asset allocation across returned, active, and overdue statuses" (and "status donut charts") with "a pie chart illustrating genre popularity and asset distribution across catalog categories."
4	High	Tab 2 / Section 4.2.3 / REQ083	Change "exclusively through Microsoft O365 institutional accounts" to "through Microsoft O365 institutional accounts (via Azure OAuth) alongside institutional email and password authentication."
5	Med	Tab 2 / Section 4.1.9 & 4.1.21 / REQ031, REQ049	Change "access a Q&A support log to assist with user troubleshooting" and "review Q&A support logs" to "manage the student Help Center FAQ knowledge base entries displayed on student dashboards."
6	Med	Tab 2 / Section 4.1.30 / REQ063	Qualify REQ063: "The student assistant shall be able to access a view-only monitoring list of system-wide borrowing and returning transactions, provided they are active and granted circulation management permissions."
7	Med	Tab 2 / Section 4.1.32 / REQ065	Qualify REQ065: "The student assistant shall be able to record student check-ins and check-outs with timestamps by scanning library cards, provided they have been granted attendance management permissions."
8	Med	Tab 2 / Section 4.1.9 & 4.1.21 / REQ028, REQ049	State the exact policy defaults and limits (14 days loan period, 5 books borrow limit, 7 days hold pickup window, 3 active holds limit, 1 day due reminder); explicitly clarify that Lumina manages borrow durations and limits without monetary fine calculation.
9	Med	Tab 2 / Section 4.1.6 / REQ019	Change "view line graphs representing daily traffic distributions and circulation trends" to "view a bar chart of peak visiting hours and an area/line trend chart of circulation volume over time."
10	Med	Tab 2 / Section 4.1.27 / REQ059	Update REQ059: "The student assistant shall be able to browse catalog inventory cards and inspect copy-level records (including copy status and generated book QR codes) in a view-only detail modal."

#	Severity	Where in document	Problem and fix
11	Low	Tab 2 / Section 4.1.37 Title	Rename section heading to "4.1.37. Catalog (Student)".
12	Low	Tab 2 / Section 4.1.21 / REQ049	Clarify in REQ049 that backend lifecycle maintenance tools (annual account reset, automated maintenance dispatch, and test email execution) are restricted to the Super Admin role.

3. Tabs 3-8 (20)

#	Severity	Where in document	Problem and fix
1	High	Tab 4 / Specific Problem / Super Admin / Analytics Module (lines 69-76)	State that Analytics is accessible to both Super Admin and Librarian; correct visualizations to: Peak Visiting Hours bar chart, Circulation Trends line graph, Genre Popularity category distribution donut chart, and High-Demand Catalog leaderboard; remove claim of asset status breakdown donut chart.
2	High	Tab 4 / Specific Problem / Super Admin / User Directory (lines 78-83)	State that the module manages accounts across all roles (students, student assistants, staff, administrators) and is shared by Super Administrators and Librarians.
3	High	Tab 4 / Specific Problem / Super Admin / Attendance Logs Module (lines 84-89)	Update role scope to indicate that scanning kiosk and system logs are operated by staff (Super Admin, Librarian, and Student Assistants with attendance permission), while students have personal attendance history access.
4	High	Tab 4 / Specific Problem / Student Assistant / Centralized Dashboard (lines 194-207)	Remove the claim of a standalone centralized inventory monitoring dashboard for Student Assistants; describe the Student Assistant dashboard as a student dashboard augmented with dynamic staff desk quick shortcuts based on role permissions.
5	High	Tab 4 / Specific Problem / Student Assistant / Inventory Module (lines 215-220)	Revise student assistant inventory capabilities to read-only collection browsing/searching, removing claims that assistants can create books or execute automated ISBN fetches.
6	High	Tab 4 / Specific Problem / Student Assistant / Borrowing History (lines 229-235)	Replace the administrative borrowing logs text with the student personal borrowing history description (tracking personal active loans, return deadlines, and overdue items).
7	High	Tab 6 / Result and Discussion / Testing (lines 376-385)	Add an empirical testing results subsection detailing automated test coverage, including test suite structure, test suite count (51 files), total automated test cases (1,017 unit/integration tests), 100% pass rate, and execution methodology.
8	High	Tab 6 / Result and Discussion (lines 376-418)	Insert an ISO/IEC 25010 evaluation section in Tab 6 presenting quantitative evaluation criteria, instrument metrics, test verification results, or Likert survey data mapping to the 8 standard quality characteristics.
9	High	Tab 7 / Abstract (line 423)	Insert the full research title "Lumina: A Web-Based Library Book Reservation, Borrowing, and Inventory Management System with QR-Based Scanning for STI College Alabang" after "Title of research:".
10	High	Tab 8 / Conclusion and Recommendation / Recommendation 2 (lines 476-482)	Reframe Recommendation 2 to focus specifically on automated institutional accreditation metric transformations (e.g., student-to-book ratio calculations, retention rates, automated PII scrubbing for CHEd/accreditation reporting) rather than building basic CSV/JSON exporters that already exist.
11	Med	Tab 4 / Specific Problem / Super Admin / Setting and Policies Module (lines 90-97)	Clarify shared access between Super Admin and Librarian for core policy configuration, specifying that backend lifecycle maintenance tools are exclusive to Super Admin.
12	Med	Tab 4 / Specific Problem / Super Admin & Librarian / Login Module (lines 46-49, 123-127, 188-192, 252-256)	Clarify that Microsoft Entra ID (O365) OAuth is the primary SSO mechanism, complemented by institutional email domain gating and credential authentication fallback.
13	Med	Tab 4 / Specific Problem / Super Admin / My Profile Module (lines 114-120)	Update profile description to mention user avatar photo upload and cropping, while identity and enrollment fields remain view-only.
14	Med	Tab 4 / Specific Problem / Librarian / Centralized Dashboard (lines 128-132)	Expand dashboard description to include the multi-metric staff KPI cards, overdue/pending action centers, and live circulation ticker rather than describing it solely as an inventory stock monitor.

#	Severity	Where in document	Problem and fix
15	Med	Tab 4 / Specific Problem / Student / Catalog Module (lines 272-277)	Explicitly mention the distinct student catalog interface and its integrated online reservation/hold placement capability.
16	Med	Tab 5 / Preliminary Design (lines 333-335)	Update technological stack description from generic "HTML, CSS, TSX" to Next.js 16 App Router, React 19, Tailwind CSS, Radix UI, and Supabase SSR.
17	Low	Tab 3 / Calendar of Activities (line 35)	Correct spelling to "Methodology".
18	Low	Tab 5 / Design of Software (lines 337-373)	Replace raw omission placeholders with numbered, captioned figure references and UI screen layout descriptions for each user persona.
19	Low	Tab 6 / Description of Prototype (lines 392-394)	Note both client-side QR scanning (`html5-qrcode`) and server-side QR generation/rasterization (`qrcode`, `sharp`).
20	Low	Tab 7 / Abstract (lines 433, 440, 441)	Revise sentences: "Lumina... is a system that addresses crucial aspects..."; "establish a widely accessible and secure platform"; "web-based application framework".

4. Tabs 9-11 (I/O, flowcharts, tables) (20)

#	Severity	Where in document	Problem and fix
1	High	Tab 9 / Input Table / 1. Student Information (lines 8-26)	Replace "Program" with "Department" and remove "Year Level" (or note it is derived/omitted); document actual profile fields (`department`, `phone`, `address`, `status`).
2	High	Tab 9 / Input Table / 2. Book Information (lines 27-49)	Clarify that books store metadata (`title`, `author`, `isbn`, `dewey_decimal`, `published_year`, `category_id`, `location`, `section`), while stock and availability are determined by `total_copies`, `available_copies`, and copy-level status in `book_copies`.
3	High	Tab 9 / Input Table / 3. Reservation Information (lines 50-65)	Change "Reservation Status" valid values from "Pending" to "Active / Ready / Fulfilled / Cancelled / Expired"; replace "BK-001" with UUID/accession identifier; add `queue_position` and `hold_expires_at` fields.
4	High	Tab 9 / Input Table / 4. Borrowing and Return Information (lines 66-87)	Update inputs to reflect `book_copy_id` (scanned copy QR string `QR-<number>` / accession number) and student library card (`LIB-...`); add staff audit fields (`processed_by`, `returned_by`) and `BorrowStatus` enum values.
5	High	Tab 9 / Input Table (lines 4-106) & Output Table (lines 108-148)	Add an "Attendance Information" input subsection (Student Card QR / Student ID, Scan Timestamp, Action: Check-in/Check-out, Notes) and corresponding output entries (Attendance Confirmation, Attendance Log Table).
6	High	Tab 9 / Report Specifications (lines 149-183)	Reframe Report Specifications to match actual system exports: Catalog Export (Excel/CSV/JSON/ZIP covers), Audit Log Export (CSV/JSON/XLSX), and interactive web dashboard analytics/logs (Borrowing History/Logs, Attendance Logs).
7	High	Tab 11 / TAB 11: Tables (lines 286-343)	Add the complete Data Dictionary / Database Schema tables to Tab 11 detailing table names, column names, data types, constraints (PK, FK, unique, check), default values, and relationships matching `schema_clean_install.sql`.
8	High	Tab 11 / Table 3 Minimum Hardware Requirements (For Development) (lines 324-326)	Move "3.7GHz (6 cores, 12 threads)" to the RECOMMENDED Processor row for AMD Ryzen 5 5600X; leave RECOMMENDED Memory as "16GB DDR4".
9	High	Tab 11 / Table 4 Minimum Hardware Requirements (For Users) (lines 340-341)	Fix Graphics row to specify a display adapter / WebGL-capable browser graphics (e.g. "Integrated graphics with DirectX 10 / WebGL support") and move or delete the sound card text.
10	Med	Tab 9 / Input Table / Examples (lines 17, 56, 72, 94)	Standardize Student ID examples to the official 11-digit STI format (`02000123456`) across all tables; replace "BK-001" with Accession Number (`ACC-00001`) or Book UUID.
11	Med	Tab 9 / Output Table / QR Scan Result (lines 139-141)	Update description to specify the 3 scanning contexts: Library Card identification, physical Book Copy checkout/checkin, and Attendance kiosk logging.

#	Severity	Where in document	Problem and fix
12	Med	Tab 10 / Figure List (lines 199-202)	Renumber `Figure 4.4 Super-Admin Borrowing Logs Flowchart` to `Figure 4.3`, and renumber subsequent Super-Admin figures (Analytics to 4.4, User Directory to 4.5, Attendance Logs to 4.6, Settings & Policies to 4.7, Audit Logs to 4.8, Profile to 4.9, Notifications to 4.10).
13	Med	Tab 10 / Figures 6.6 & 7.2 (lines 262, 278)	Rename captions to "Figure 6.6 Student-Assistant My Borrowing Flowchart" and "Figure 7.2 Student My Borrowing Flowchart".
14	Med	Tab 10 / Figures 6.1, 6.3, 6.5 (lines 250, 254, 260)	Add a note to the flowchart descriptions clarifying that Student-Assistant access to Inventory, Circulation Desk, and Attendance Logs is conditioned on active status and delegated admin permissions.
15	Med	Tab 10 / Librarian Figures (lines 222-245)	Keep Audit Logs out of Librarian flowcharts; add an explicit explanatory note confirming that Audit Logs is restricted exclusively to Super-Admin by design.
16	Med	Tab 11 / Table 2 Software Requirements (lines 308-310)	Update Table 2 to state: Database: "PostgreSQL (hosted on Supabase)"; Technologies: "Next.js 15, TypeScript, Tailwind CSS, shadcn/ui, Zod, Supabase Auth & Storage".
17	Low	Tab 10 / Figure 4.8 (line 213)	Change "Setting & Policies" to "Settings & Policies".
18	Low	Tab 10 / Figure 5.5 (line 235)	Change "LIbrarian" to "Librarian".
19	Low	Tab 10 / Figure 6.7 (line 264)	Change to "Figure 6.7 Student-Assistant Borrowing Logs Flowchart".
20	Low	Tab 11 / Table 2 Software Requirements (line 312)	Change "Microsoft 0365 Word" to "Microsoft Office 365 Word" or "Microsoft 365 Word".

5. User Guide (Tab 12) (22)

#	Severity	Where in document	Problem and fix
1	High	Tab 12 / 2. User Roles and Features (lines 24-25, 60), 13. Quick Reference Table (lines 223-231)	Expand role documentation into 4 distinct roles (`student`, `student_assistant`, `librarian`, `super_admin`); document Student Assistant's dual views and delegated permissions, and distinguish Librarian from Super Admin.
2	High	Tab 12 / 2. Key Features (line 33), 9. Common Functions / A. Reporting a Missing Book (lines 174-180), 13. Quick Reference Table (line 228)	Remove Section 9.A "Reporting a Missing Book", and remove "Report missing books" from Key Features and the Quick Reference Table.
3	High	Tab 12 / 2. Key Features / How To: Reserve a Book (lines 47-51)	Update reservation instructions: state that the button reads "Reserve for Pickup" (or "Reserve to Borrow" for waitlisted titles), and placing a hold is immediate upon click with toast notification. Document how students can cancel existing holds.
4	High	Tab 12 / 3. Circulation Desk Functions / A. Checking Out a Book (lines 99-106)	Detail the 3-step wizard workflow (Identify Student -> Scan Resource -> Review); replace "Select Process" with "Click 'Confirm Checkout'".
5	High	Tab 12 / 3. Circulation Desk Functions / B. Returning a Book (lines 107-113)	Detail the 2-step Return workflow (Identify Resource -> Review); replace "Select Process" with "Click 'Confirm Return'".
6	High	Tab 12 / 5. Managing Reservations (lines 133-140)	Clarify that staff monitor reservation queues inside the Inventory book details modal (`ReservationQueuePanel`), and process/fulfill them at the Circulation Desk (`/circulation`).
7	High	Tab 12 / 8. System Settings and Policies (lines 155-173)	Rename module to "Settings & Policies" (`/policies`); remove "Renewal settings"; list actual setting categories (Circulation, Reservations, Identity, Categories, Broadcasts, Support, Lifecycle); and document the 2-step audit commit process ("Review Changes" -> reason min 10 chars -> "Confirm & Commit").
8	High	Tab 12 / 10. Updating Personal Information (lines 191-197)	State that student personal information is read-only and synchronized from institutional records; document that the only permitted profile update is uploading and cropping an avatar photo via Avatar Manager.

#	Severity	Where in document	Problem and fix
9	High	Tab 12 / 13. Quick Reference Table for Roles and Access (lines 223-231)	Restructure the Quick Reference Table to list all 4 roles (`student`, `student_assistant`, `librarian`, `super_admin`) against all primary modules (Dashboard, Catalog, Inventory, Circulation, Borrowing Logs, Attendance Logs, Analytics, User Directory, Settings & Policies, Audit Logs).
10	Med	Tab 12 / 1. Logging Into the System / 1.2 Microsoft 365 Login (line 18)	Change "Sign in with Microsoft 365" to "Sign in with Microsoft"; mention the Quick Demo Access buttons available for thesis evaluation and testing.
11	Med	Tab 12 / 2. Key Features / View Active Borrowed Books (lines 52-55)	Rename "Active Borrows section" to "Current Borrows section" in the student dashboard guide.
12	Med	Tab 12 / 2. Key Features / How To: Search for a Book (line 43), Reserve a Book (line 48)	Note that the "Catalog" (`/student-catalog`) is student-facing, while library personnel use the "Inventory" module (`/inventory`) to search and inspect books.
13	Med	Tab 12 / 2. Key Features (lines 36, 56-58), 6. Borrow History (lines 141-147), 13. Quick Reference Table (line 227)	Replace static "Borrow History" with dynamic labels: "My Borrowing" for students and "Borrowing Logs" for library personnel (noting both are available to authorized student assistants).
14	Med	Tab 12 / 2. Key Features (lines 37-38), 9. Common Functions / B. Viewing the Library Card (lines 181-185)	Document the "Front" / "Back" card flip toggle, the enlarged QR modal popup for circulation kiosk scanning, and clarify that "Export" downloads a high-resolution PNG image.
15	Med	Tab 12 / 2. Key Features (line 39), 9. Common Functions / C. Viewing FAQs (lines 186-190)	Clarify that FAQs appear directly on the Student Dashboard in the "Support & FAQs" section, displaying accordion-style questions and answers without navigating away.
16	Med	Tab 12 / 2. Key Features / How To: Add a Book (lines 79-83)	Update navigation path to "Inventory" (`/inventory`), describe the ISBN auto-lookup capability, and update the submit button label to "Save to Catalog".
17	Med	Tab 12 / 2. Key Features / How To: Edit a Book (lines 84-89)	Update flow: open Inventory, click book card, click the edit icon (pencil) in the modal header, update metadata, and click "Save Changes".
18	Med	Tab 12 / 2. Key Features / How To: Generate or Print a QR Code (lines 90-94)	Specify that QR codes are generated per physical copy; update modal title to "Print QR Label" and action button to "Print Label".
19	Med	Tab 12 / 7. User Directory (lines 148-154)	Document specific features: tab filters (notably "Pending Review" and "Archived"), card approval workflow, bulk re-activation button, and permission assignment for Student Assistants.
20	Med	Tab 12 / 12. Troubleshooting and Support (lines 204-222)	Add specific troubleshooting entries for "Account Pending Approval", "Account Suspended", "Camera Access Blocked" (use manual input), and "Reservation Limit Reached".
21	Med	Tab 12 / 2. Key Features (lines 80, 85), 4. Managing Inventory (line 128)	Replace "Admin Dashboard" with "Dashboard" for metrics/action center tasks, and with "Inventory" (`/inventory`) for book management, addition, and editing tasks.
22	Low	Tab 12 / 11. Logging Out (lines 198-203), 3. Circulation Desk Functions / C. Resetting the Scanner (line 115)	Update label to "Log out", document the logout confirmation dialog, and mention the scanner camera Stop/Switch controls alongside wizard Reset.